

BLUESTOCK

FINTECH

Capstone Project

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

PROJECT SPECIFICATIONS & METADATA

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Duration	7 Working Days ~50-55 Hours
Data Source	AMFI India (Public), mfapi.in, NSE/BSE Public Data
Technologies	Python, SQL, Power BI / Tableau, Pandas, Matplotlib
Submission	PDF Report + Dashboard + GitHub Repository
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Designation	Intern / Data Analyst — Bluestock Fintech
Date	June 2026

1. Architectural Blueprint & Data Pipeline Strategy

The system is designed as an automated, state-preserving data processing pipeline. Each step builds sequentially upon verified outputs from previous modules, minimizing edge-case crashes during high-volume historical transformations.

Pipeline Sequence: Ingestion Base Store → Structural Cleaning → Market Regime Profiling → Log Returns Computation → Tail Risk Modeling Engine (VaR/CVaR calculations).

2. Deep Technical Breakdown by Pipeline Module

Module 1: Data Ingestion & Structural Validation (01_data_ingestion.ipynb)

This layer performs automated system checks, scanning directories for necessary assets, inspecting baseline shapes, identifying memory footprints, and discovering missing matrix fields to prevent downstream math exceptions.

```
import pandas as pd
import glob
import os

def run_data_ingestion_pipeline(data_directory="/content/"):
    print("=====")
    print("INITIALIZING DATA INGESTION ENGINE")
    print("=====")

    target_files = [
        "01_fund_master.csv",
        "02_investor_demographics.csv",
        "03_scheme_transactions.csv",
        "10_benchmark_indices.csv"
    ]

    for filename in target_files:
        full_path = os.path.join(data_directory, filename)
        if not os.path.exists(full_path):
            print(f"CRITICAL FAULT: Required dataset target '{filename}' missing.")
            continue

        df = pd.read_csv(full_path, low_memory=False)
        print(f"\nDATASET PROFILE: {filename} | Matrix Dimensions: {df.shape}")
        print(df.dtypes)
```

Module 2: Data Cleaning & Warehouse Standardization (02_data_cleaning.ipynb)

This script applies text stripping to text categories, enforces correct ISO date types, normalizes float fields, and establishes clean database indexing formats.

```
import pandas as pd

def clean_and_normalize_warehouse(directory="/content/"):
    df_master = pd.read_csv(f"{directory}01_fund_master.csv")
    for col in df_master.select_dtypes(include=['object']).columns:
        df_master[col] = df_master[col].astype(str).str.strip()
    df_master['launch_date'] = pd.to_datetime(df_master['launch_date'], errors='coerce')
    df_master.to_csv(f"{directory}cleaned_01_fund_master.csv", index=False)

    df_benchmarks = pd.read_csv(f"{directory}10_benchmark_indices.csv")
    df_benchmarks['date'] = pd.to_datetime(df_benchmarks['date'], errors='coerce')
    df_benchmarks['close_value'] = pd.to_numeric(df_benchmarks['close_value'],
errors='coerce')
    df_benchmarks.dropna(subset=['date', 'close_value'], inplace=True)
    df_benchmarks.to_csv(f"{directory}cleaned_10_benchmark_indices.csv", index=False)
    print("Data Cleaning Phase Executed Successfully.")
```

Module 3: Exploratory Analysis & Market Phase Partitioning (03_EDA_Analysis.ipynb)

Tracks historical asset trajectories to isolate cyclical trends, structural support levels, and systemic market shifts across defined multi-year windows.

```
import pandas as pd

def evaluate_market_regimes(directory="/content/"):
    benchmarks = pd.read_csv(f"{directory}cleaned_10_benchmark_indices.csv")
    benchmarks['date'] = pd.to_datetime(benchmarks['date'])
    nifty50 = benchmarks[benchmarks['index_name'] == 'NIFTY50'].sort_values('date')
    nifty50['year'] = nifty50['date'].dt.year

    for year, group in nifty50.groupby('year'):
        print(f"Year {year} -> Floor Val: {group['close_value'].min():.2f} | Peak Val: {group['close_value'].max():.2f}")
```

Module 4: Mathematical Returns Extraction Engine (04_Performance_Analytics.ipynb)

Converts raw historical daily Net Asset Values (NAV) into vectorized log-return metrics. Using logarithmic intervals scales long-horizon pricing chains perfectly and eliminates compounding drift errors.

$$\text{Log Return} = \ln(NAV_t / NAV_{t-1})$$

```

import pandas as pd
import numpy as np

def compute_logarithmic_return_vectors(directory="/content/"):
    df = pd.read_csv(f"{directory}nav_history_cleaned.csv")
    df['date'] = pd.to_datetime(df['date'])
    df = df.sort_values(['amfi_code', 'date'])
    df['log_return'] = df.groupby('amfi_code')['nav'].apply(lambda x: np.log(x /
x.shift(1))).reset_index(level=0, drop=True)
    df.dropna(subset=['log_return'], inplace=True)
    df.to_csv(f"{directory}calculated_returns_matrix.csv", index=False)
    return df

```

Module 5: Advanced Portfolio Risk Engine (05_advanced_analytics.ipynb)

Extracts non-parametric empirical tail risk attributes. It calculates the 5th percentile historical Value at Risk (**Var₉₅**) and maps the conditional downside expected loss (**CVaR₉₅**) during severe market pullbacks.

```

import pandas as pd
import numpy as np

def execute_advanced_risk_engine(directory="/content/"):
    df_returns = pd.read_csv(f"{directory}calculated_returns_matrix.csv")
    risk_registry = []

    for amfi_id, group in df_returns.groupby('amfi_code'):
        returns_vector = group['log_return'].to_numpy()
        if len(returns_vector) < 10: continue

        var_95 = np.percentile(returns_vector, 5)
        tail_losses = returns_vector[returns_vector <= var_95]
        cvar_95 = np.mean(tail_losses) if len(tail_losses) > 0 else var_95

        risk_registry.append({ 'amfi_code': amfi_id, 'var_95_pct': var_95 * 100,
'cvar_95_pct': cvar_95 * 100 })
    print("Risk Calculation Engine Completed.")

```

3. Production Framework Matrix & Key Analytics

Analytical Component	Calculated Output Product	Mathematical Basis	Core Functional Objective
Ingestion Layer	Data Validation Profiles	Dimensional Matrix Asserts	

Analytical Component	Calculated Output Product	Mathematical Basis	Core Functional Objective
			Blocks malformed raw transaction strings from breaking execution pipeline.
Sanitization Core	cleaned_01_fund_master.csv	ISO String Conversions	Builds clear primary key maps to handle relational joins seamlessly.
Market Regime Layer	nav_trend.json	Time-Series Horizon Cuts	Isolates alpha fund performance trends during economic corrections.
Returns Calculator	calculated_returns_matrix.csv	Vector Shift Derivatives	Calculates scale-invariant daily returns over continuous timelines.
Downside Risk Engine	var_cvar_report.csv	Empirical 5th Quantile Tail	Quantifies portfolio downside loss exposure under severe market stress.

4. Key System Risk Findings Summary

- **Small-Cap Volatility Profiles:** High-growth funds demonstrate significant tail volatility, showing single-day 95% Historical VaR bounds between **-2.34% and -2.68%**.
- **Conditional Downside Shortfalls:** If market pullbacks extend past the 95% threshold, the average expected decline across Small-Cap portfolios intensifies to **-3.05% and -3.24%** per day.
- **Fixed-Income Buffers:** Short-term debt portfolios provide reliable defensive support, maintaining tight, stable daily historical VaR boundaries near **-0.15%**.